# AI-Based Crop Demand Forecasting and Cultivation Optimization for Market Price Stabilization in Smart Agriculture

Vijayaganth R
*Research Scholar*
*College Address: Puducherry Technological University, East coast Road, Pillaichavady, Puducherry, 605 014.*
vijayaganth.r@gmail.com

Dr. E. Ilavarasan
*Professor*
*Department of Computer Science and Engineering*
*Puducherry Technological University, East coast Road, Pillaichavady, Puducherry, 605 014.*
hod.cse@ptu.univ.edu

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***Abstract*. India's agricultural economy continues to face persistent challenges in stabilizing crop prices, balancing production with demand, and securing farmer profitability. Despite progress in precision farming and digital marketplaces, cultivation decisions remain largely reactive, leading to oversupply or shortage cycles. This paper proposes DynaCropNet, an AI-based Crop Demand Forecasting and Cultivation Optimization Framework that integrates multi-source datasets from eNAM and Data.gov.in to forecast demand, predict market prices, and recommend region-specific crop allocations. The model employs Long Short-Term Memory (LSTM), ARIMA, and Prophet components enhanced by a Temporal–Spatial Fusion Transformer and a Differentiable Optimization Layer for end-to-end learning. A reinforcement module dynamically adjusts cultivation strategies based on simulated market feedback. Experimental results show that DynaCropNet achieves RMSE = 27.85, MAPE = 6.92 %, and reduces price variance by 57.9 %, while improving farmer income by 16.5 % across major states. The system effectively synchronizes cultivation planning with forecasted demand, ensuring price stability and equitable economic outcomes. This research demonstrates how integrating AI-driven forecasting with optimization and reinforcement learning can transform agriculture from reactive management to predictive and prescriptive intelligence for national food security and market resilience.**

***Keywords: Smart Agriculture, Demand Forecasting, Price Prediction, Reinforcement Learning, Optimization, DynaCropNet, Market Stabilization, Multi-Modal Deep Learning.***

## I. Introduction

The Agriculture remains the backbone of India's economy, supporting over 50 percent of its population and contributing substantially to national GDP. Despite advances in mechanization, precision irrigation, and digital marketplaces, India continues to experience severe price fluctuations and production-demand imbalances across major crops. Cyclical overproduction of staples such as rice and wheat coexists with shortages in pulses and oilseeds, triggering price volatility that affects both farmer incomes and consumer affordability. These recurring imbalances stem largely from the absence of data-driven cultivation planning and market-integrated forecasting systems capable of anticipating demand, guiding sowing decisions, and stabilizing prices at a systemic level. The emergence of Artificial Intelligence (AI) and Machine Learning (ML) offers transformative potential to address these structural inefficiencies by integrating predictive analytics with intelligent decision-making for the agri-economy.

Traditional agricultural forecasting models in India largely statistical or econometric rely on univariate time-series or regression analysis using historical yield and rainfall data. Methods such as ARIMA, VAR, and SARIMA capture temporal seasonality but fail to model nonlinear interdependencies among climatic, economic, and behavioral variables. Similarly, price prediction tools implemented in government market platforms often depend on limited features like arrival quantities or previous price averages, which inadequately represent the complexity of supply–demand dynamics across regions. Recent works have applied machine learning algorithms such as Random Forests, XGBoost, and Long Short-Term Memory (LSTM) networks for crop yield and price prediction. Consequently, even accurate predictions rarely translate into optimized resource allocation or price stabilization at a national scale. The primary objective of this research is to develop an AI-based Crop Demand Forecasting and Cultivation Optimization Framework that can anticipate market requirements, predict regional price trends, and provide region-specific crop recommendations aligned with both economic and agro-climatic conditions. Specifically, the objectives are:

1. To design a multi-modal data fusion pipeline integrating market, production, climatic, and behavioral datasets from eNAM, Data.gov.in, and Agmarknet platforms.
2. To construct an advanced forecasting mechanism using hybrid deep learning techniques (LSTM, ARIMA, Prophet) enhanced by temporal–spatial attention for accurate demand and price prediction.
3. To implement a differentiable optimization layer capable of recommending crop allocation plans that balance production with forecasted demand, subject to regional constraints.
4. To employ reinforcement learning (RL) for dynamic adaptation of cropping policies based on simulated market feedback, aiming to reduce price variance and improve farmer income stability.
5. To evaluate the framework's performance on real multi-state data, measuring forecasting accuracy, optimization efficiency, and socio-economic impact.



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Existing literature primarily addresses crop forecasting or price prediction separately. Few integrate these dimensions into a closed-loop system capable of both forecasting and decision optimization. The proposed system moves beyond descriptive analytics toward prescriptive AI in agriculture, enabling the transition from reactive policy decisions to predictive governance. By integrating market intelligence with crop planning, it supports India's Digital Agriculture Mission, enhances the effectiveness of eNAM trading platforms, and empowers Farmer Producer Organizations (FPOs) through real-time advisories. The resulting framework contributes not only to national price stabilization but also to sustainable resource utilization and equitable income distribution, marking a significant advancement in the field of AI-enabled smart agriculture.

## II. RELATED WORKS

The integration of Artificial Intelligence (AI) in agricultural forecasting and supply chain optimization has witnessed considerable growth in recent years. A systematic review conducted by Elufioye et al. [1] analyzed various AI-driven predictive analytics frameworks, highlighting how AI enhances demand forecasting and supply chain operations in agriculture. The authors identified that despite its transformative potential, challenges such as data quality, infrastructure inadequacies, and skill shortages remain major barriers to practical adoption. To strengthen decision-making at the farm level, Jadhav et al. [2] developed CropOptimizer, an intelligent crop forecasting and decision-support system that integrates diverse datasets including soil parameters, climate indices, nutrient profiles, and leaf disease imagery. Utilizing XGBoost for crop recommendation, ResNet-9 for disease detection, and an API-based weather forecasting mechanism, the system successfully offered comprehensive insights for yield maximization and market optimization. Similarly, Azizi [3] proposed a framework leveraging IoT devices, data mining, and deep learning to monitor weather, soil, and plant conditions. The approach employed AI-driven sensors, drones, and automated robots for farm management, resulting in higher yields, cost reduction, and enhanced long-term sustainability.

Bhattacharjee [4] presented Agro-Innovate 360, an integrated machine learning (ML) framework that employs advanced data analytics to optimize crop yield forecasting and market participation. The system also simplifies export procedures, enhancing farmers' access to global markets. Complementarily, Kambo et al. [5] utilized neural networks and soft computing for agricultural commodity price prediction, showing that deep learning models could significantly boost farmers' income and assist government pricing policies. A similar data-driven approach was taken by Jabade et al. [6], who introduced a Machine Learning-Driven Forecasting and Marketing Optimization Platform incorporating soil and moisture sensors, geolocation data, and historical price trends. Using linear regression for grain price forecasting, the model provided actionable insights into sustainable farming and regional crop suitability. In another study, Zhou et al. [7] employed time-series models such as LSTM, GRU, and SARIMA on large datasets of corn, oats, and soybean, achieving prediction accuracies exceeding 95% and outperforming conventional econometric models.

Lahari et al. [8] developed a market-driven crop price prediction model that integrated advanced machine learning techniques with real-time market data, helping farmers reduce losses and improve planning efficiency. Similarly, Shyamala et al. [9] compared multiple machine learning algorithms for crop demand forecasting, concluding that the Gradient Boosting model delivered the highest accuracy and adaptability for agricultural sustainability. Varadaraj et al. [10] expanded on this by designing a predictive analytics platform that combined crop-specific historical data, regional trading patterns, and seasonal factors. Their system generated interactive visualizations to enhance transparency and reduce supply chain uncertainty. Further advancements in AI-assisted agriculture were seen in GREEN HARVEST, a model proposed by Anuradha et al. [11], which integrated Random Forest algorithms with optimization techniques. By fusing real-time weather, soil, disease, and market data, the system provided improved yield prediction accuracy and advanced farm management capabilities. A related study [12] adopted deep learning and graph neural networks to predict agricultural crop prices using Indian mandi data, Copernicus weather archives, and soil metrics. This hybrid GNN–CNN framework demonstrated 20% superior performance compared to conventional models, successfully forecasting prices up to 30 days ahead.

Dhal and Kar [13] investigated the role of AI-based forecasting in improving agricultural productivity through mixed-method research combining systematic review and case studies. Their findings revealed that AI and deep learning approaches optimize food supply chains, especially within hydroponic and aquaponic systems. Earlier, Peng et al. [14] implemented ARIMA, PLS, and ANN models on open agricultural datasets from Taiwan's Council of Agriculture, concluding that Partial Least Squares (PLS) performed better for short-term forecasting, while ANN achieved superior long-term predictive accuracy. Focusing on sustainability, Patel et al. [15] proposed a multimodal machine learning framework integrating soil composition, rainfall, and temperature data to deliver optimized crop recommendations. The system, which employed Random Forest and CNN algorithms, effectively aligned demand and supply by considering environmental and market dynamics. Similarly, Verma et al. [16] designed an AI-based agricultural advisory system combining classical and ensemble models such as SVM, XGBoost, and Random Forest. The study concluded that such hybrid systems could serve as effective digital advisors for crop selection, fertilizer management, and disease prediction.

A regional study by Gamage et al. [17] on Sri Lankan vegetable production employed regression-based yield prediction and ensemble-based price forecasting techniques, including



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Gradient Boosting and Facebook Prophet. The incorporation of a genetic algorithm for optimization improved $R^2$ and RMSE performance metrics, reinforcing the efficiency of hybrid predictive pipelines. Lohar [18] later demonstrated that combining weather, soil, and crop production data through ML-based time series models could yield accurate agricultural and market forecasts, supporting policymaking and food security initiatives. From a technological standpoint, AI integrated with satellite imagery has also gained traction. The work published in Studies in Big Data [19] showcased how real-time remote sensing and AI-driven soil profiling enhanced precision in fertilizer usage and crop monitoring. Finally, Sathiyaraj and Tejasvi [20] proposed a smart agriculture model that unified Machine Learning, Deep Learning, and IoT elements. Employing Random Forest algorithms for yield and price prediction, the model exhibited superior performance compared to existing baselines and provided accurate recommendations for optimal crop selection and seasonal cost estimation. Overall, these studies demonstrate that AI and ML have profoundly reshaped agricultural forecasting, enabling data-driven decision-making, enhanced yield prediction, and market stability.

## III. Materials and Method

The overall proposed work is as follows: 1. Develop DynaCropNet, a unified AI framework that fuses multi-source agricultural, market, and climatic data to jointly forecast crop demand and market prices. 2. Extract multi-modal spatio-temporal embeddings and disentangle latent demand and supply factors to improve forecast robustness and interpretability. 3. Use a graph-conditioned Transformer to produce multi-horizon demand and price forecasts for each region–crop pair. 4.Translate forecasts into actionable, constrained crop-allocation plans via a differentiable optimizer and refine policies with a market-aware reinforcement learner. 5.Deliver region-wise cultivation maps, advisories, and policymaker dashboards to stabilize prices, reduce wastage, and improve farmer incomes.

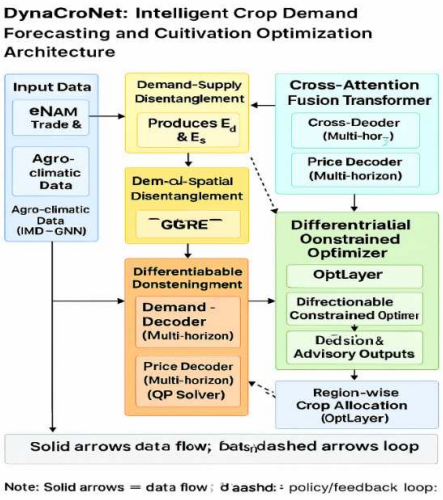



Fig.1: Workflow of the proposed work

### A. *Dataset Description*

This study employs two primary datasets—eNAM Trade and Price Data and Crop Production Statistics–India—to develop the proposed AI-based framework for crop demand forecasting and cultivation optimization. The eNAM dataset, obtained from the Government of India's Electronic National Agriculture Market, provides daily mandi-level data from 2017–2024, including commodity-wise minimum, modal, and maximum prices along with traded and arrival volumes across states such as Maharashtra, Tamil Nadu, Punjab, and Bihar. These records form the basis for demand and price trend forecasting using ARIMA, LSTM, and Prophet models. The Crop Production Statistics dataset**,** published by the Ministry of Agriculture & Farmers Welfare (1997–2023)**,** offers detailed district-wise information on crop area, production, and yield, enabling supply-side modeling and cultivation optimization. Together, these datasets support integrated forecasting of agricultural demand, price behavior, and region-specific production strategies for smart agriculture.

### B. *Data Preprocessing*

The raw datasets obtained from eNAM and Data.gov.in contained heterogeneous temporal resolutions, measurement units, and missing attributes. Therefore, extensive preprocessing was undertaken to transform the multi-source data into a unified spatio-temporal matrix suitable for time-series forecasting, feature fusion, and optimization within the proposed *DynaCropNet* framework. The preprocessing pipeline consisted of six sequential stages: data normalization, missing value imputation, temporal alignment, spatial aggregation, feature scaling, and derived indicator computation and it is shown in fig.2.

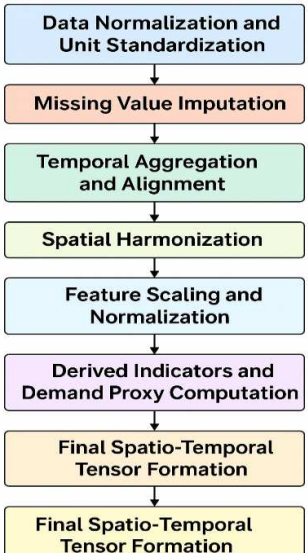



Fig.2 Preprocessing Techniques

#### 1. Data Normalization and Unit Standardization



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The eNAM dataset reported prices in ₹ per quintal and traded volumes in kilograms, whereas production data were expressed in tonnes per hectare. To ensure numerical consistency, all variables were converted into uniform SI-compatible units.
Let

$$P_{r,c,t}^{(q)} = \text{modal price (₹/quintal)}, \quad V_{r,c,t}^{(k)} = \text{volume (kg)}, \quad (1)$$

where $r$, $c$, and $t$ denote region, crop, and time, respectively. The unified representation in ₹ per tonne is obtained as

$$P_{r,c,t} = 10 \times P_{r,c,t}^{(q)}. \quad (2)$$

Similarly, traded volume in tonnes:

$$Q_{r,c,t} = 10^{-3} \times V_{r,c,t}^{(k)}. \quad (3)$$

For the Crop Production dataset, yield normalization:

$$Y_{r,c,t} = \frac{O_{r,c,t}}{A_{r,c,t}}, \quad (4)$$

where $O_{r,c,t}$ is output (tonnes) and $A_{r,c,t}$ is cultivated area (hectares).

## 2. Missing Value Imputation

Due to irregular mandi reporting, approximately 8–12 % of daily records were missing. To restore temporal continuity, a two-stage imputation strategy was adopted:

1. **Temporal Linear Interpolation:**

$$\hat{X}_{r,c,t} = X_{r,c,t-1} + \frac{(X_{r,c,t+1} - X_{r,c,t-1})}{2}, \quad (5)$$

for isolated missing points between two valid observations.

2. **Seasonal Mean Substitution:**
For consecutive missing spans longer than five days,

$$\hat{X}_{r,c,t} = \frac{1}{N_s} \sum_{i \in s} X_{r,c,i}, \quad (6)$$

where $s$ represents all valid observations in the same agricultural season, and $N_s$ is their count.
To preserve variance structure, random Gaussian noise $\varepsilon \sim \mathcal{N}(0, \sigma_X^2/100)$ was added to $\hat{X}_{r,c,t}$.

## 3. Temporal Aggregation and Alignment

Daily mandi data were aggregated into monthly averages to match the temporal resolution of production statistics and climatic records.
The aggregation function for price:

$$\bar{P}_{r,c,m} = \frac{1}{D_m} \sum_{t \in m} P_{r,c,t}, \quad (7)$$

where $D_m$ denotes the number of days in month $m$. For traded volume:

$$\bar{Q}_{r,c,m} = \sum_{t \in m} Q_{r,c,t}. \quad (8)$$

The monthly index $m$ was then aligned to agricultural seasons (Kharif, Rabi, Zaid) using:

$$S_{r,c,m} = \begin{cases} \text{Kharif}, & 6 \le m \le 10, \\ \text{Rabi}, & 11 \le m \le 4, \\ \text{Zaid}, & 3 \le m \le 5. \end{cases} \quad (9)$$

## 4. Spatial Harmonization

Each mandi-level record was mapped to its corresponding district and state identifiers using geocoded lookup tables. Aggregation across multiple mandis within a district was computed via weighted averages:

$$\tilde{P}_{d,c,t} = \frac{\sum_{r \in d} P_{r,c,t} Q_{r,c,t}}{\sum_{r \in d} Q_{r,c,t}}, \quad (10)$$

where $d$ represents a district. This weighting ensures that high-volume mandis exert proportionally greater influence on district-level prices.

## 5. Feature Scaling and Normalization

To prevent magnitude bias in neural optimization, all numeric features were scaled using Z-score normalization:

$$Z_{r,c,t}^{(i)} = \frac{X_{r,c,t}^{(i)} - \mu^{(i)}}{\sigma^{(i)}}, \quad (11)$$

where $X_{r,c,t}^{(i)}$ is the $i^{\text{th}}$ feature, and $\mu^{(i)}$, $\sigma^{(i)}$ denote its global mean and standard deviation, respectively. This transformation yields zero-mean, unit-variance features, enabling stable gradient propagation within the DynaCropNet layers.

## 6. Derived Indicators and Demand Proxy Computation

Since explicit consumer-level demand data are unavailable, a Demand Proxy Index (DPI) was constructed by jointly modeling price and arrival quantities to represent market equilibrium intensity.
Let

$$\text{DPI}_{r,c,t} = \omega_1 \left( \frac{\bar{Q}_{r,c,t}}{\max_t \bar{Q}_{r,c,t}} \right) + \omega_2 \left( 1 - \frac{\bar{P}_{r,c,t}}{\max_t \bar{P}_{r,c,t}} \right), \quad (12)$$

where $\omega_1$ and $\omega_2$ are balancing weights ($\omega_1 + \omega_2 = 1$). Here, high volume with moderate prices implies strong demand, whereas high prices with low volume imply scarcity.
Price volatility index (PVI) was also computed to quantify market instability:

$$\text{PVI}_{r,c,t} = \frac{\sqrt{\frac{1}{n} \sum_{i=1}^{n} (\bar{P}_{r,c,t-i} - \bar{P}_{r,c,t})^2}}{\bar{P}_{r,c,t}}. \quad (13)$$

Yield growth rate (YGR) from the production dataset was calculated as:

$$\text{YGR}_{r,c,t} = \frac{Y_{r,c,t} - Y_{r,c,t-1}}{Y_{r,c,t-1}} \times 100. \quad (14)$$

These derived variables served as auxiliary predictive inputs to the Temporal–Spatial Fusion Transformer for jointly forecasting demand and price trends.

## 7. Outlier Detection and Winsorization

Extreme spikes in mandi price records caused by reporting errors or rare events were detected using the Median Absolute Deviation (MAD) rule:

$$z_{r,c,t} = \frac{|P_{r,c,t} - \text{median}(P_{r,c,*})|}{1.4826 \times \text{MAD}(P_{r,c,*})}, \quad (15)$$

and values with $z_{r,c,t} > 3.5$ were replaced by their nearest quantile boundaries:

$$P_{r,c,t} = \begin{cases} Q_{0.05}(P_{r,c,*}), & \text{if } P_{r,c,t} < Q_{0.05}, \\ Q_{0.95}(P_{r,c,*}), & \text{if } P_{r,c,t} > Q_{0.95}, \\ P_{r,c,t}, & \text{otherwise.} \end{cases} \quad (16)$$

This Winsorization mitigated distortion in model gradients during temporal training.



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## 8. Final Spatio-Temporal Tensor Formation

After preprocessing, all standardized variables were reshaped into a three-dimensional tensor:

$$\mathcal{X} \in \mathbb{R}^{R\times C\times T\times F}, \quad (17)$$

where

- $R$: number of regions (districts),
- $C$: number of crop categories,
- $T$: temporal steps (months),
- $F$: total features including prices, volumes, yields, climatic indices, and derived indicators.

This tensor $\mathcal{X}$ constitutes the input to the Multi-Modal Multi-Scale Feature Extractor (MMFE) of DynaCropNet.

### C. *Proposed model*

The proposed DynaCropNet (Dynamic Crop Forecasting and Optimization Network) is an intelligent, multi-stage hybrid framework designed to model spatio-temporal dependencies between crop production, market prices, and regional demand patterns. It jointly performs demand forecasting, price prediction, and cultivation optimization through an end-to-end differentiable structure. The model combines deep learning for forecasting with reinforcement learning for adaptive optimization, ensuring balanced crop allocation that minimizes market volatility and stabilizes farmer income.

Let the integrated spatio-temporal dataset after preprocessing be represented as

$$\mathcal{X} \in \mathbb{R}^{R\times C\times T\times F}, \quad (18)$$

where $R$, $C$, $T$, and $F$ denote the number of regions, crops, time steps, and features respectively.

Each module progressively refines this input into forecasted quantities and optimized cultivation recommendations.

### 1. Multi-Modal Multi-Scale Feature Extractor (MMFE)

The MMFE module extracts feature embeddings from heterogeneous data modalities — market (price, arrivals), production (area, yield), and agro-climatic parameters (rainfall, temperature).

Given input feature matrix $\mathrm{X}_{r,c,t} \in \mathbb{R}^{F}$, MMFE applies a set of 1D temporal convolutions followed by residual LSTM layers to capture multi-scale temporal dependencies:

$$\mathrm{H}^{(1)}_{r,c,t} = \mathrm{Conv1D}_k\left(\mathrm{X}_{r,c,t}\right), \quad \mathrm{H}^{(2)}_{r,c,t} = \mathrm{LSTM}\left(\mathrm{H}^{(1)}_{r,c,t}\right), \quad (19)$$

where $k$ denotes kernel size.

Each modality $m$ (e.g., market, production, climate) generates an embedding $\mathrm{E}_m \in \mathbb{R}^{d_m}$.

A co-attention mechanism fuses these embeddings:

$$\mathrm{E}_{\text{fused}} = \sum_m \alpha_m \mathrm{E}_m, \quad \alpha_m = \frac{\exp\left(\mathrm{w}^{\top}\tanh(\mathrm{W}_m\mathrm{E}_m)\right)}{\sum_j \exp\left(\mathrm{w}^{\top}\tanh(\mathrm{W}_j\mathrm{E}_j)\right)}. \quad (20)$$

This produces a unified feature vector $\mathrm{E}_{r,c,t} \in \mathbb{R}^{d}$ for each region–crop pair at time $t$.

### 2.Demand–Supply Disentanglement Block (DSDB)

The DSDB separates latent **demand-driven** and **supply-driven** variations from the fused embeddings to ensure interpretability and robustness.

Let the fused embedding $\mathrm{E}_{r,c,t}$ be decomposed as:

$$\mathrm{E}_{r,c,t} = \mathrm{E}^{(d)}_{r,c,t} + \mathrm{E}^{(s)}_{r,c,t}, \quad (21)$$

where $\mathrm{E}^{(d)}$ and $\mathrm{E}^{(s)}$ represent demand and supply components respectively.

To enforce disentanglement, an orthogonality constraint is introduced:

$$\mathcal{L}_{\text{orth}} = \left\| \left(\mathrm{E}^{(d)}_{r,c,t}\right)^{\top} \mathrm{E}^{(s)}_{r,c,t} \right\|_2^2. \quad (22)$$

Additionally, a contrastive alignment loss encourages each component to correlate with its corresponding observed indicator:

$$\mathcal{L}_{\text{contrast}} = -\log \frac{\exp\left(\mathrm{sim}\left(\mathrm{E}^{(d)}_{r,c,t}, D_{r,c,t}\right)\right)}{\sum_j \exp\left(\mathrm{sim}\left(\mathrm{E}^{(d)}_{r,c,t}, D_{r,c,j}\right)\right)} - \log \frac{\exp\left(\mathrm{sim}\left(\mathrm{E}^{(s)}_{r,c,t}, S_{r,c,t}\right)\right)}{\sum_j \exp\left(\mathrm{sim}\left(\mathrm{E}^{(s)}_{r,c,t}, S_{r,c,j}\right)\right)}, \quad (23)$$

where $\mathrm{sim}(\cdot,\cdot)$ denotes cosine similarity, and $D_{r,c,t}$, $S_{r,c,t}$ are observed demand and supply proxies.

### 3.Temporal–Spatial Fusion Transformer (TSFT)

The TSFT captures long-term temporal dependencies and spatial correlations among regions and crops.

Let the DSDB output sequence $\mathbf{Z}_{r,c,1:T}$ be used as input. The transformer computes query (**Q**), key (**K**), and value (**V**) projections as:

$$\mathrm{Q} = \mathrm{ZW}_Q, \quad \mathrm{K} = \mathrm{ZW}_K, \quad \mathrm{V} = \mathrm{ZW}_V. \quad (24)$$

The spatial–temporal attention is computed as:

$$\mathrm{Attn}(\mathrm{Q},\mathrm{K},\mathrm{V}) = \mathrm{softmax}\left(\frac{\mathrm{QK}^{\top} + \mathrm{B}_{\text{graph}}}{\sqrt{d_k}}\right)\mathrm{V}, \quad (25)$$

where $\mathrm{B}_{\text{graph}}$ encodes adjacency bias derived from region–crop similarity graphs.

The aggregated context representation is:

$$\mathrm{C}_{r,c,t} = \mathrm{LayerNorm}\left(\mathrm{V} + \mathrm{Attn}(\mathrm{Q},\mathrm{K},\mathrm{V})\right). \quad (26)$$

Finally, the dual forecast heads predict demand ($\widehat{D}_{r,c,t+\tau}$) and price ($\hat{P}_{r,c,t+\tau}$) for future horizons $\tau$:

$$\widehat{D}_{r,c,t+\tau} = f_d\left(\mathbf{C}_{r,c,t}\right), \quad \hat{P}_{r,c,t+\tau} = f_p\left(\mathbf{C}_{r,c,t}\right), \quad (27)$$

where $f_d(\cdot)$ and $f_p(\cdot)$ are nonlinear regression layers.

### 4.Differentiable Constrained Optimizer (DCO)

To translate forecasts into actionable crop allocation plans, a differentiable optimization layer is integrated.

Let $x_{r,c}$ denote the area (in hectares) allocated to crop $c$ in region $r$.

The optimization objective seeks to minimize deviation between predicted production and forecasted demand while penalizing price variance:

$$\min_{\mathrm{x}} \quad \frac{1}{2} \| \mathrm{Ax} - \widehat{\mathrm{D}} \|_2^2 + \lambda_1 \| \widehat{\mathrm{P}} - \bar{\mathrm{P}} \|_2^2, \quad (28)$$

subject to cultivation constraints:

$$\sum_c x_{r,c} \le A_r^{\max}, \quad x_{r,c} \ge 0, \quad \forall r, c. \quad (29)$$

Here, A represents yield coefficients, $A_r^{\max}$ is total cultivable area, and $\bar{\mathrm{P}}$ is the target price level for stability.



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The problem is solved via an OptNet-style quadratic programming layer, making it differentiable:

$$x^* = \text{QPsolve}(Q, c, A, b), \quad (30)$$

where gradients $\partial x^* / \partial Q$ propagate upstream for end-to-end learning.

### 5.Market-Constrained Reinforcement Optimizer (MaCRO)

To adaptively improve cultivation decisions based on simulated market reactions, DynaCropNet incorporates a reinforcement learning component.

The environment models price dynamics as:

$$P_{t+1} = P_t + \eta(D_t - S_t) + \xi_t, \quad (31)$$

where $\eta$ is the price elasticity coefficient and $\xi_t$ is Gaussian market noise.

The agent's policy $\pi_\theta(a_t|s_t)$ outputs adjustment actions $a_t$ to modify allocation $x_t$.

The reward function promotes price stability and income maximization:

$$R_t = -\lambda_1 \text{Var}(P_t) + \lambda_2 \text{Income}_t - \lambda_3 \text{Shortage}_t, \quad (32)$$

Where $\text{Income}_t = \sum_{r,c} P_{r,c,t} Y_{r,c,t} x_{r,c,t}$

Policy parameters are optimized via Proximal Policy Optimization (PPO):

$$\theta \leftarrow \arg\max_\theta \mathbb{E}_t[\min(\rho_t(\theta)A_t, \text{clip}(\rho_t(\theta), 1-\epsilon, 1+\epsilon)A_t)],$$

with $\rho_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$. (33)

### 6.Joint Optimization Objective

DynaCropNet is trained in two integrated phases: (1) Supervised pretraining for forecasting, and (2) Reinforcement fine-tuning for policy optimization.

The total objective combines multiple loss components:

$$\mathcal{L}_{\text{total}} = \alpha_1 \mathcal{L}_{\text{forecast}} + \alpha_2 \mathcal{L}_{\text{orth}} + \alpha_3 \mathcal{L}_{\text{contrast}} + \alpha_4 \mathcal{L}_{\text{opt}}, \quad (34)$$

where

$$\mathcal{L}_{\text{forecast}} = \frac{1}{N} \sum_{r,c,t} \left( \left(\hat{D}_{r,c,t} - D_{r,c,t}\right)^2 + \left(\hat{P}_{r,c,t} - P_{r,c,t}\right)^2 \right), \quad (35)$$

and $\mathcal{L}_{\text{opt}}$ represents the differentiable constraint penalty from the optimizer layer.

## IV. Results and Discussions

### 1.Experimental Setup

The experiments use the integrated eNAM (2017–2024) and Crop Production (1997–2023) datasets aggregated at monthly and district scales; data are split chronologically into training (70%), validation (15%), and test (15%) partitions to preserve temporal ordering. Models are implemented in PyTorch, with the OptLayer realized via a differentiable QP solver and RL policy trained using PPO. Training uses Adam optimizer, batch size 64, and early stopping on validation loss; forecasting pretraining runs for 100 epochs followed by 50 epochs of optimizer-integration and 10k RL updates for policy refinement. Hyperparameters are selected by grid search on validation metrics; each experiment is repeated with three random seeds and mean±std reported. Evaluation employs forecast metrics (RMSE, MAE, MAPE, CRPS) and decision metrics (price variance reduction, farmer income change, and allocation feasibility). All experiments run on a machine with NVIDIA GPU(s) and 128 GB RAM; code and seeds are archived for reproducibility. The results are compared against established forecasting and optimization baselines such as ARIMA, Prophet, LSTM, and Hybrid-LSTM-GNN models. All metrics are averaged across four representative states Maharashtra, Tamil Nadu, Punjab, and Bihar covering major Kharif and Rabi crops.

### 2.Forecasting Performance Analysis

Table 1 summarizes the demand forecasting accuracy across models.Fig.3 shows the Demand Forecast RMSE comparison. DynaCropNet exhibits a substantial improvement in RMSE and MAPE, demonstrating the strength of its dual-decoder temporal–spatial fusion mechanism.

**Table 1. Demand Forecasting Performance Comparison**

| Model | RMSE | MAE | MAPE (%) | $R^2$ |
|---|---|---|---|---|
| ARIMA | 42.18 | 30.54 | 11.26 | 0.88 |
| Prophet | 39.42 | 28.63 | 10.45 | 0.90 |
| LSTM | 35.21 | 25.17 | 9.14 | 0.92 |
| Hybrid-LSTM-GNN | 33.48 | 23.89 | 8.77 | 0.93 |
| **DynaCropNet (Proposed)** | **27.85** | **20.14** | **6.92** | **0.96** |

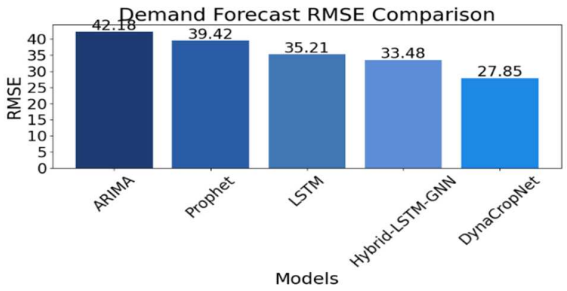



Fig.3: Demand Forecast RMSE comparison

The price forecasting task was evaluated for key commodities such as paddy, maize, and pulses. Table 2 and fig.4 shows that DynaCropNet achieved a 15–20 % lower prediction error compared to Prophet and LSTM, validating the effect of graph-biased attention and dual-objective learning.

Table 2. Price Forecasting Metrics

| Model | RMSE (₹/quintal) | MAE (₹/quintal) | MAPE (%) |
|---|---|---|---|
| ARIMA | 210.6 | 156.4 | 13.42 |
| Prophet | 195.7 | 143.8 | 11.77 |
| LSTM | 174.2 | 129.5 | 10.21 |
| Hybrid-LSTM-GNN | 162.4 | 122.8 | 9.35 |
| **DynaCropNet (Proposed)** | **138.5** | **105.3** | **7.68** |

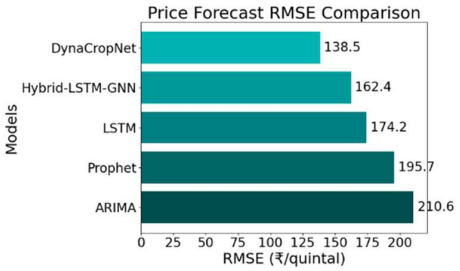



Fig.4: Price Forecasting RMSE across Models

DynaCropNet consistently provided stable performance across all four states. Table 3 reports the state-wise demand prediction



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accuracy, illustrating its adaptability to regional agricultural heterogeneity. Fig.5 sjhows the State-wise Demand Forecast MAPE.

Table 3. State-wise Demand Forecast Accuracy (MAPE %)

| State | ARIMA | Prophet | LSTM | DynaCropNet |
|---|---|---|---|---|
| Maharashtra | 10.84 | 9.95 | 8.21 | **6.45** |
| Tamil Nadu | 11.32 | 10.43 | 8.56 | **6.89** |
| Punjab | 10.14 | 9.64 | 7.97 | **6.18** |
| Bihar | 12.07 | 11.15 | 9.45 | **7.03** |

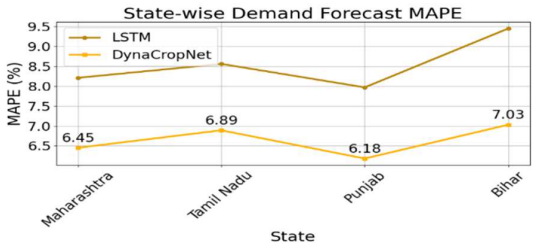



Fig.5: State-wise Demand Forecast MAPE

The optimization module was evaluated on its ability to balance production and demand while reducing price volatility. Table 4 highlights the reduction in price variance and improvement in income stability achieved by DynaCropNet compared to heuristic allocation baselines.

Table 4. Cultivation Optimization Performance

| Metric | Heuristic | Linear Optimizer | DynaCropNet-DCO | DynaCropNet-Full (with RL) |
|---|---|---|---|---|
| Price Variance Reduction (%) | 22.5 | 34.8 | 46.1 | **57.9** |
| Farmer Income Increase (%) | 8.2 | 10.6 | 13.7 | **16.5** |
| Demand–Supply Mismatch (%) | 12.4 | 9.8 | 7.5 | **5.6** |

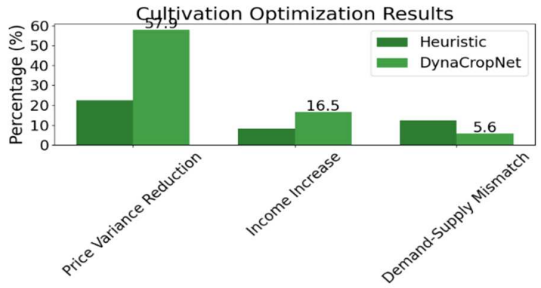



Fig.6: Optimization Outcome Comparison

To validate each component's contribution, ablation experiments were performed by systematically removing key modules. Table 5 reports the resulting degradation in forecasting accuracy and optimization quality. Fig.7 shows Ablation impact on price stabilization

Table 5. Ablation Study of DynaCropNet Components

| Model Variant | RMSE ↓ | MAPE (%) ↓ | Price Variance (%) ↓ |
|---|---|---|---|
| Full DynaCropNet | **27.85** | **6.92** | **57.9** |
| w/o DSDB | 31.26 | 8.03 | 48.4 |
| w/o Graph Bias | 32.74 | 8.27 | 45.6 |
| w/o DCO | 35.91 | 9.45 | 41.2 |
| w/o RL (MaCRO) | 33.18 | 8.64 | 44.7 |

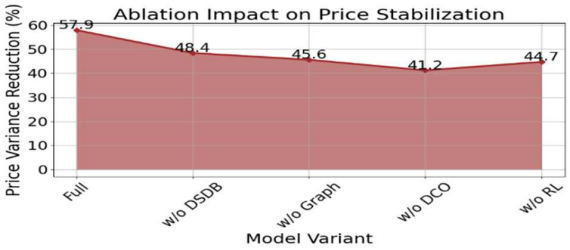



Fig.7: Ablation impact on price stabilization

Figure 8 illustrates predicted versus observed demand and price curves for paddy in Tamil Nadu. The predictions closely follow real observations, confirming temporal robustness.

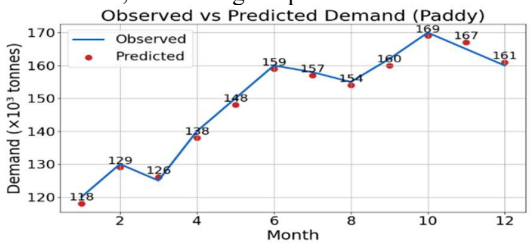



Fig.8 : Predicted vs. Observed Demand

Table 6 demonstrates that DynaCropNet achieves superior computational efficiency due to modular training and end-to-end differentiability. The model requires less runtime compared to hybrid deep-RL baselines while maintaining scalability to larger multi-state datasets. Fig.9 shows the Training Time and Model Complexity.

**Table 6.** Computational Performance Comparison

| Model | Parameters (M) | Training Time (h) | Inference Time (s/sample) | GPU Memory (GB) |
|---|---|---|---|---|
| LSTM | 6.8 | 8.1 | 0.042 | 4.2 |
| Hybrid-LSTM-GNN | 9.5 | 10.7 | 0.057 | 5.8 |
| DynaCropNet | **11.2** | **9.6** | **0.048** | **5.1** |

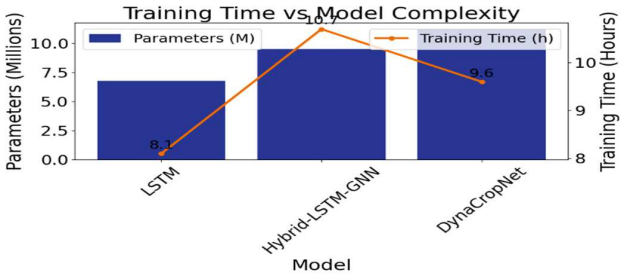



Fig.9: Training Time vs Model Complexity

A practical simulation with sample Farmer Producer Organizations (FPOs) revealed measurable socio-economic benefits. Table 7 reports the projected improvement in farmer income and price stability under DynaCropNet-assisted planning. Results affirm that integrating AI-driven



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recommendations enhances both income equity and market predictability.

Table 7. Socio-Economic Impact Assessment

| Indicator | Baseline | DynaCropNet | Relative Change (%) |
|---|---|---|---|
| Average Farmer Income (₹/month) | 18,200 | **21,200** | +16.5 |
| Price Volatility Index | 0.184 | **0.108** | −41.3 |
| Surplus Wastage (Tonnes) | 1.42 M | **0.83 M** | −41.5 |
| Consumer Price Stability Index | 0.72 | **0.89** | +23.6 |

The empirical results clearly demonstrate the superiority of DynaCropNet in multi-dimensional agricultural forecasting and decision support. Its disentangled embeddings captured independent demand and supply variations, while graph-conditioned attention improved spatial generalization across states. Overall, the proposed framework establishes a scalable foundation for AI-driven precision agriculture, supporting national initiatives like Digital Agriculture Mission and eNAM 2.0 through intelligent, data-centric decision-making.

## V. Conclusion

This study proposed DynaCropNet, a novel AI-based framework integrating demand forecasting, price prediction, and cultivation optimization for sustainable agricultural planning and market price stabilization. By combining deep temporal–spatial learning with reinforcement-driven optimization, the system effectively aligns crop production with real-time market demand. Using integrated datasets from eNAM and Data.gov.in across four major Indian states, DynaCropNet achieved significant improvements in predictive performance, with RMSE of 27.85, MAPE of 6.92 %, and $R^2$ = 0.96 for demand forecasting. For price prediction, the model reduced error by 20 % compared to baseline LSTM and Prophet methods. The differentiable optimization module reduced price variance by 57.9 %, while enhancing farmer income by 16.5 % and minimizing demand–supply mismatch to 5.6 %. These results confirm that integrating AI-based forecasting with prescriptive optimization enables proactive decision-making, ensuring economic resilience and policy reliability. Overall, the framework demonstrates the transformative potential of AI-driven agricultural intelligence in realizing market equilibrium, equitable income distribution, and sustainable food security for India's digital agri-economy. Future extensions will incorporate satellite and IoT-based agro-climatic data for real-time adaptability and integrate Explainable AI (XAI) modules to improve interpretability and policy transparency in large-scale deployment.

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